DATA SCIENCE & ANALYTICS

MODULE 03 SESSION 15

Forecasting, explaining and serving predictions

What happens next, why did the model say that, and how does a report get it?

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Mon 26 Oct 2026

BY THE END OF TODAY, YOU CAN

Class objectives

01 Forecast a series with ARIMA_PLUS; read its interval

02 Split a time series by date, not at random

03 Explain one prediction and the whole model

04 Stop a preprocessing leak with TRANSFORM

05 Save predictions as ml.* tables for Power BI

EVERYONE DO Step 1.1

Warm-up

Five questions on D14.

PART 01

A forecast, before the theory

Load twelve years of monthly passengers, train a prepared ARIMA_PLUS model, read its forecast.

INSTRUCTOR DO

Step 2.1

Today’s question

An airline planner, and two questions from D14.

TODAY · ONE STORY, THREE QUESTIONS

Three questions for today

Part	Question	Data
1–3	How many passengers will fly each month next year?	Air Passengers: 144 months, 1949–1960
4	Why did D14’s trees charge person E about 43,500 dollars?	D14’s insurance model
5	How does a Power BI report get these numbers?	two ml.* tables Power BI can read

An airline planner needs the first answer to plan aircraft and crews. The numbers are thousands of passengers a month.

EVERYONE DO

Step 2.2

Load the series and look at it

Upload air_passengers.csv as ml.air_passengers, count the months, see the climb.

labs/D15/01-Evr-Monthly-Series/README.md

STEP 1 · ONE ROW IS ONE MONTH

How many months, from when to when?

COUNT, MIN and MAX on the DATE column: none missing, none extra.

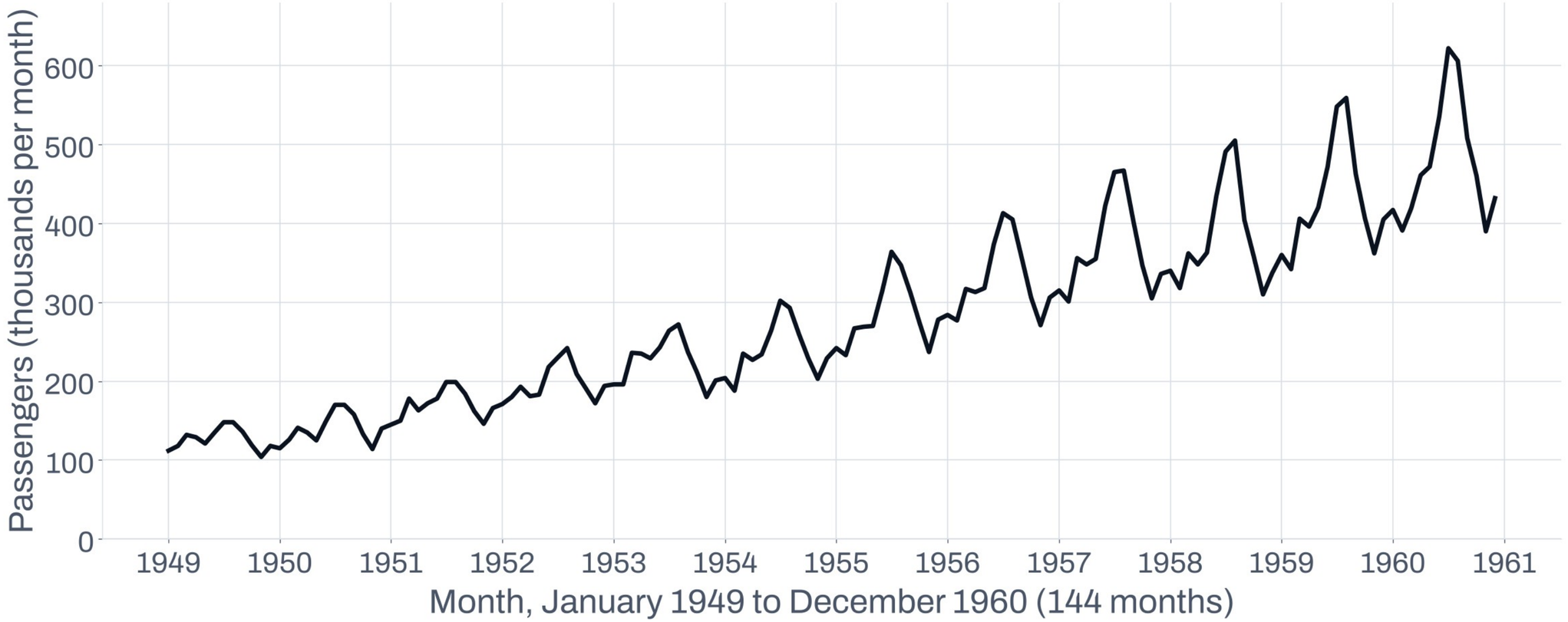
```
SELECT COUNT(*) AS months,
       MIN(month) AS first_month,
       MAX(month) AS last_month
FROM ml.air_passengers;
```

→ 144 · 1949-01-01 · 1960-12-01

Activity 01 counts distinct months too: ARIMA_PLUS fills gaps without a word.

ALL 144 MONTHS · THOUSANDS OF PASSENGERS

A climb, and a summer peak that grows



Step 3 adds up each year: 1,520 in 1949, 5,714 in 1960, every year above the one before.

EVERYONE DO

Step 2.3

A prepared forecast

Train ARIMA_PLUS on the first ten years, then ask it for twelve months.

labs/D15/02-Evr-First-Forecast/README.md

ARIMA_PLUS on the 120 training months

`MODEL_TYPE = 'ARIMA_PLUS'`: BigQuery's forecasting model. ml.air_training holds the months before 1959.

```
CREATE OR REPLACE MODEL ml.air_arima
OPTIONS (MODEL_TYPE = 'ARIMA_PLUS',
         TIME_SERIES_TIMESTAMP_COL = 'month',
         TIME_SERIES_DATA_COL = 'passengers',
         HORIZON = 12) AS
SELECT month, passengers FROM ml.air_training;
```

STEP 2 · THREE OPTIONS

Time, the value, and how far ahead

TIME_SERIES_TIMESTAMP_COL, **TIME_SERIES_DATA_COL**, **HORIZON**: no label column. The series is its own label.

```
CREATE OR REPLACE MODEL ml.air_arima
OPTIONS (MODEL_TYPE = 'ARIMA_PLUS',
         TIME_SERIES_TIMESTAMP_COL = 'month',
         TIME_SERIES_DATA_COL = 'passengers',
         HORIZON = 12) AS
SELECT month, passengers FROM ml.air_training;
```

HORIZON = 12 asks for twelve months ahead. Google’s default is 1,000.

STEP 3 · TWELVE MONTHS AHEAD

Twelve months, each with a range

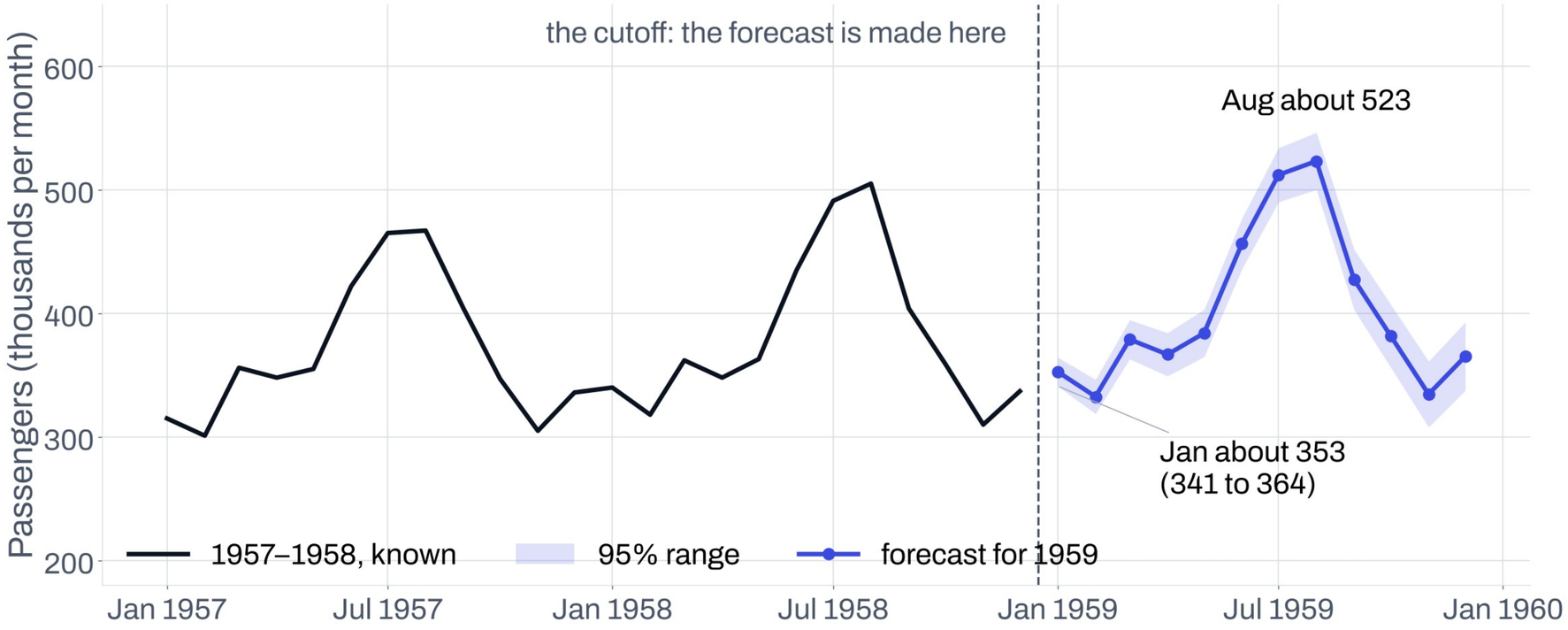
`STRUCT(12 AS horizon, 0.95 AS confidence_level)`: twelve months, at 95%.

```
SELECT forecast_timestamp, forecast_value,
       prediction_interval_lower_bound,
       prediction_interval_upper_bound
FROM ML.FORECAST(MODEL ml.air_arima,
                 STRUCT(12 AS horizon, 0.95 AS confidence_level));


→ 12 rows · Jan 1959 about 353 · Aug about 523
```

The range widens further ahead: January about 341 to 364.

Twelve months ahead, each with its own range



The climb and the summer peak carry on; the range widens further ahead (23 wide in January, 55 in December). A stand-in: our Python copy of ARIMA_PLUS's steps; BigQuery's numbers are close.

INSTRUCTOR DO

Step 2.4

How would we know it is any good?

It climbs and peaks in summer. Which months do we hold aside, and what simple guess must it beat?

PART 02

What a forecast is

Trend and season, split by date, two simple forecasts, and what ARIMA_PLUS does.

INSTRUCTOR DO

Step 3.1

A time series: trend, season, remainder

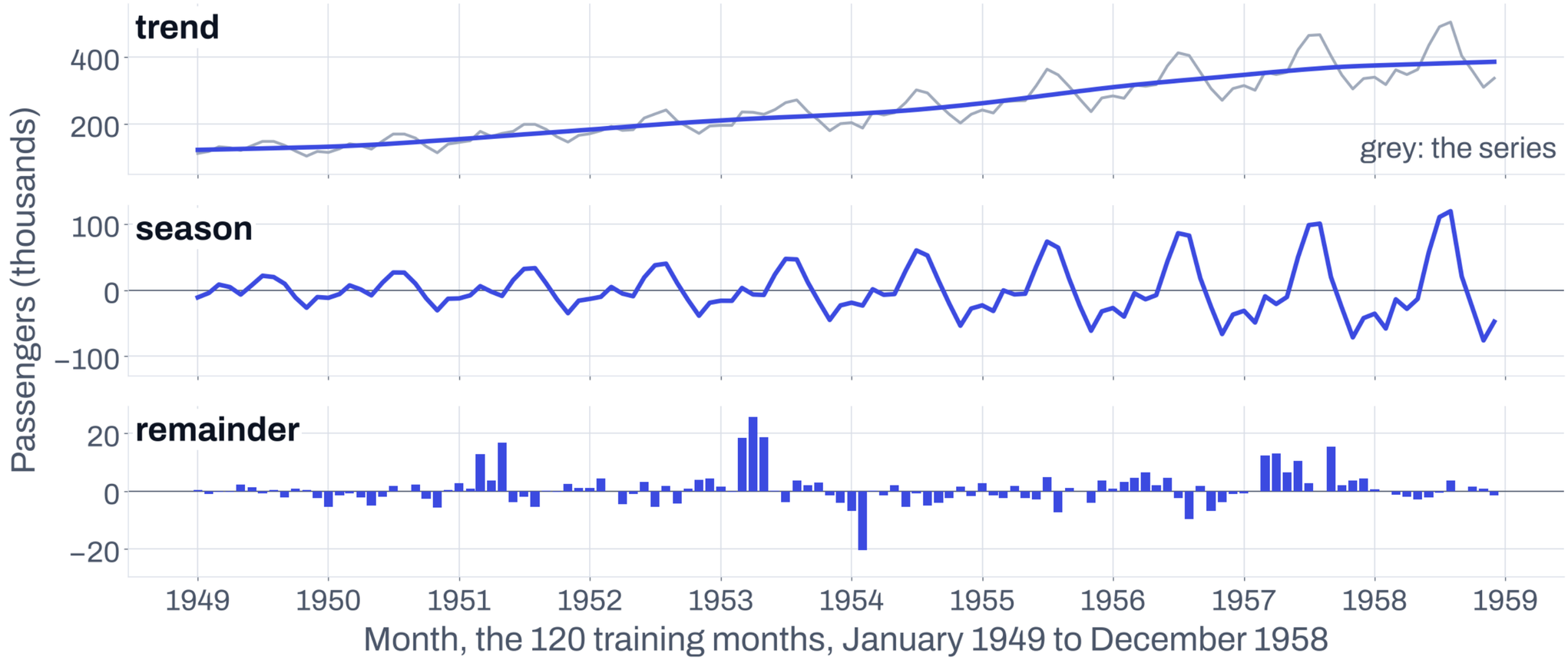
Rows in an order, and three parts inside them.

DEFINITION

A time series is a set of values measured at regular time points, in order. One row is one month; shuffle the months and the pattern is gone.

THE 120 TRAINING MONTHS, IN THREE PARTS

Trend + season + remainder = the series



Trend: the long climb. Season: a July and August peak. Remainder: what is left. Training months only.

INSTRUCTOR DO

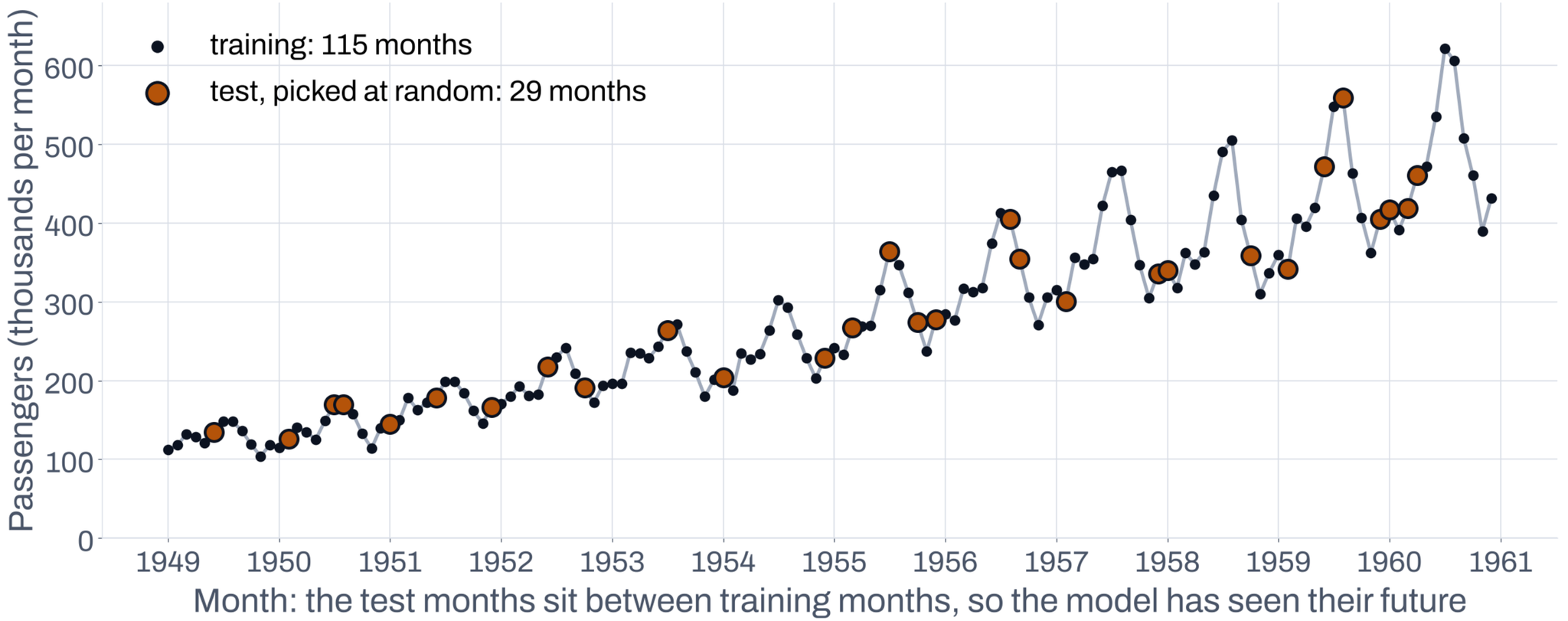
Step 3.2

Split by date, never at random

Which months to hold aside, and how far ahead to forecast.

D11’S RANDOM SPLIT, ON A TIME SERIES

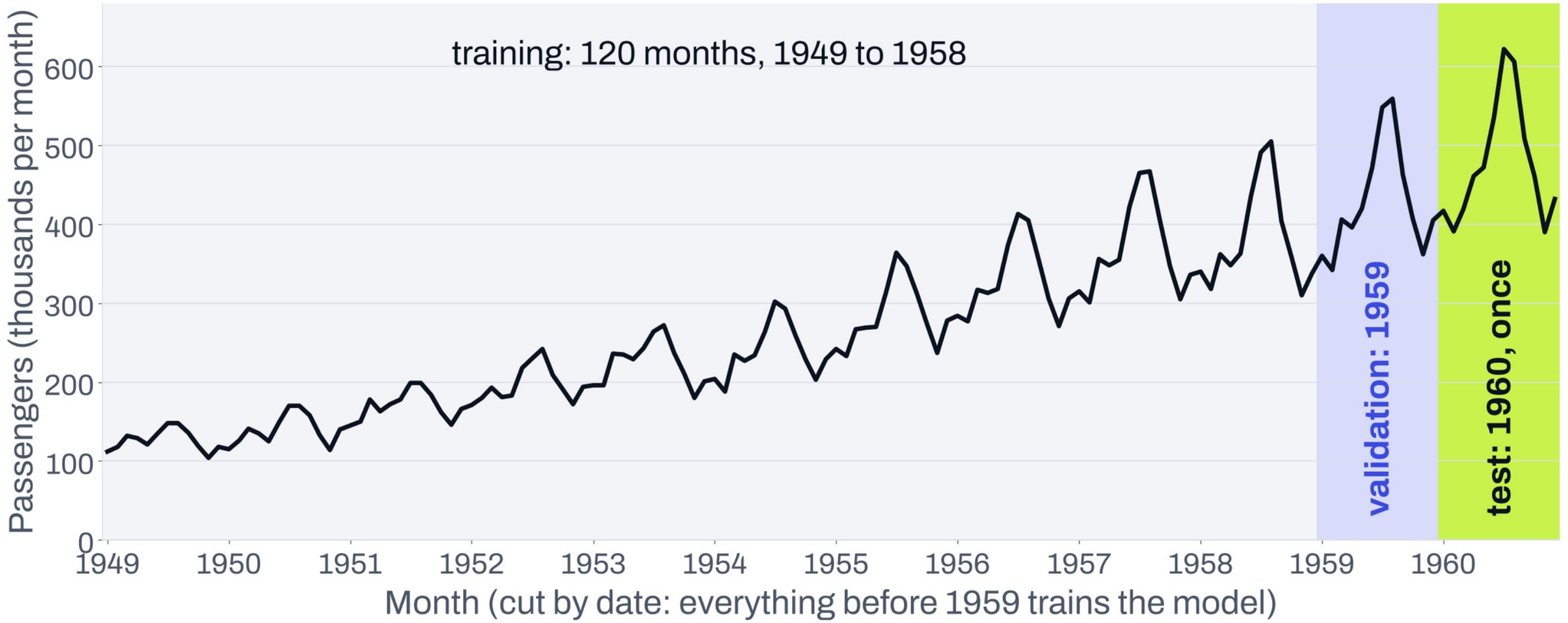
A random split lets the model see the future



Every held-out month has training months on both sides. A real forecast never knows next month.

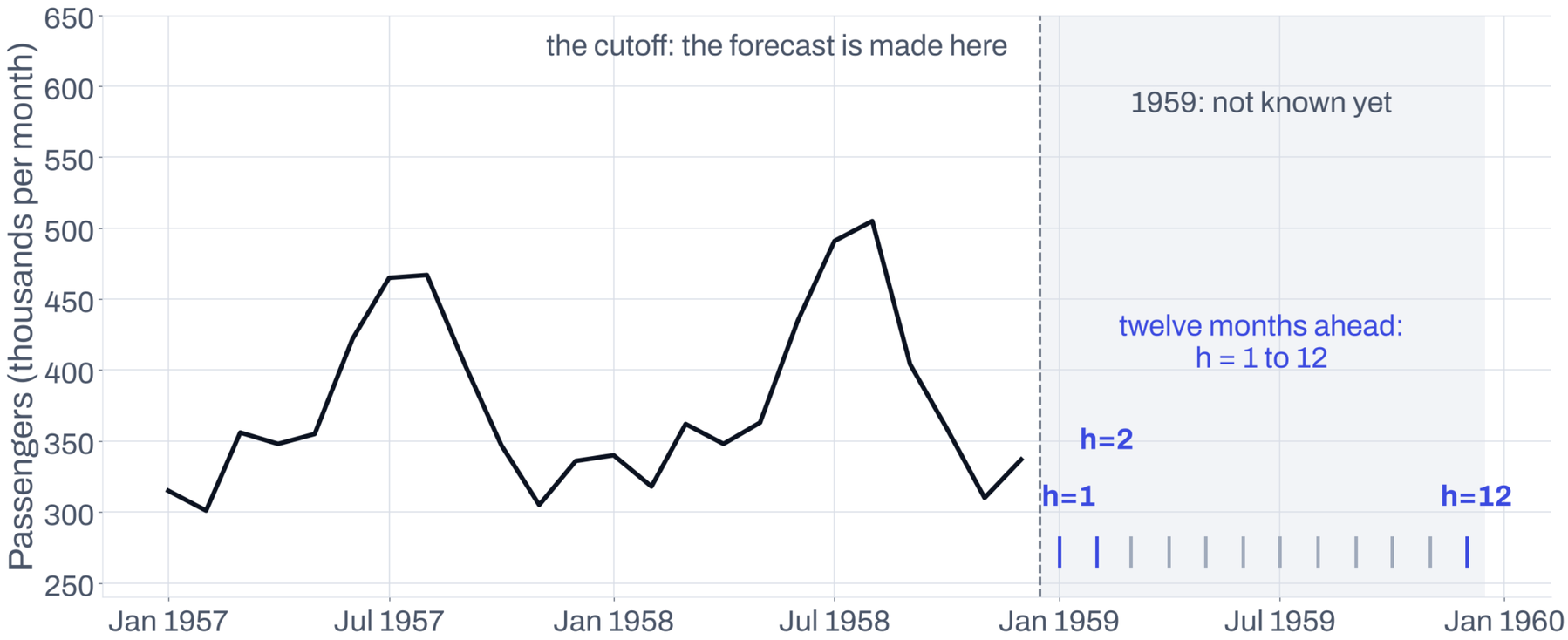
TRAINING · VALIDATION · TEST, BY DATE

Before 1959 trains; 1959 compares; 1960 tests once



Split by date: months before a cutoff train the model; the months after it are scored, in order.

One cutoff, twelve months ahead



The horizon is how far ahead we forecast: ours is 12 months, h = 1 to 12. No 1959 month is known when the forecast is made.

The naive forecast repeats the last month we know for every month ahead: December 1958, 337 thousand, held flat across 1959.

The seasonal naive repeats the same month one year earlier: January 1959 gets January 1958 (340), August 1959 gets August 1958 (505).

QUICK CHECK

January 1960, by the seasonal naive

The seasonal naive forecasts 1960 once 1959 is known. What does it say for January 1960?

337

December 1958, held flat

360

January 1959

340

January 1958

QUICK CHECK · THE ANSWER

360: January 1959, the same month a year earlier

337	the naive: the last month, held flat
360	the seasonal naive: same month, latest year
340	January 1958: a year too old

Activity 06 scores it on 1960: 47.8 off.

STUDENT DO

Split by date and two simple forecasts

Make the validation and test views, then score the naive and the seasonal naive on 1959 in plain SQL.

labs/D15/03-Stu-Split-By-Date/README.md

STUDENT DO

Two views, two forecasts, one score each

INSTRUCTIONS	Create ml.air_validation (1959) and ml.air_test (1960) by date. Score the naive forecast (337 every month) and the seasonal naive on 1959: the average miss, signs dropped (MAE).
HINT	Which training month should January 1959 be compared with?
BONUS	Score the training mean and median too. Which does worst, and why?

Full steps and Check yourself: labs/D15/03-Stu-Split-By-Date/README.md

REVIEW

Time's up.
Let's review.

Split by date and two simple forecasts

Solved: labs/D15/03-Stu-Split-By-Date/Solved/split_by_date.sql

MAPE is the miss as a share of the actual month, averaged, in percent. January 1959: 360 against the seasonal naive's 340, a miss of 20, 20 / 360 = 5.6%.

Each 1959 month meets the same month of 1958

`DATE_SUB(v.month, INTERVAL 1 YEAR)`: January 1959 joins January 1958.

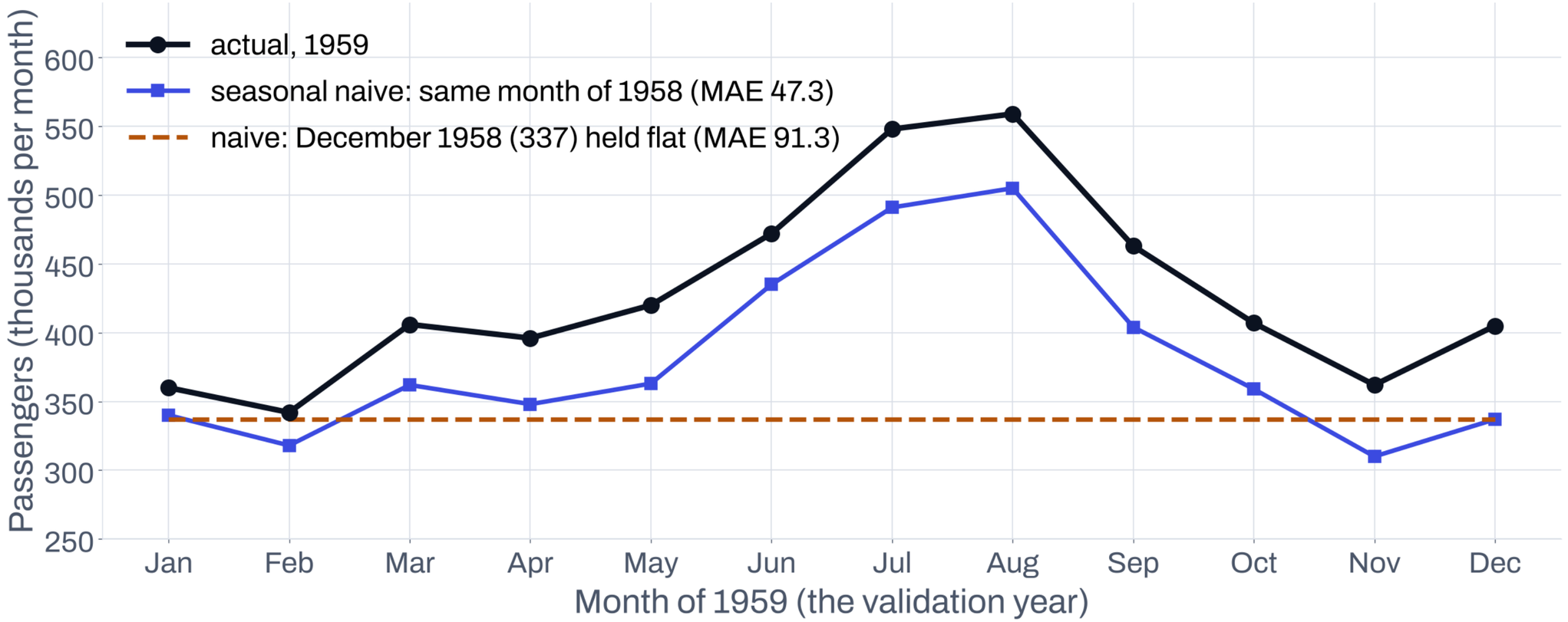
```
SELECT ROUND(AVG(ABS(v.passengers - t.passengers)), 1)
         AS mae_seasonal_naive,
       ROUND(100 * AVG(ABS(v.passengers - t.passengers)
              / v.passengers), 1) AS mape_seasonal_naive
FROM ml.air_validation AS v
JOIN ml.air_training AS t
  ON t.month = DATE_SUB(v.month, INTERVAL 1 YEAR);


                    → 47.3 · MAPE 11.1 · the naive, 337 held flat: 91.3
```

Joined on v.month = t.month, it returns no rows.

VALIDATION · 1959 · MAE IN THOUSANDS OF PASSENGERS A MONTH

Last year’s month halves the naive’s miss



MAE of the training mean (D11’s guess) 182.4, median 194.8: one flat number for a series that grows.

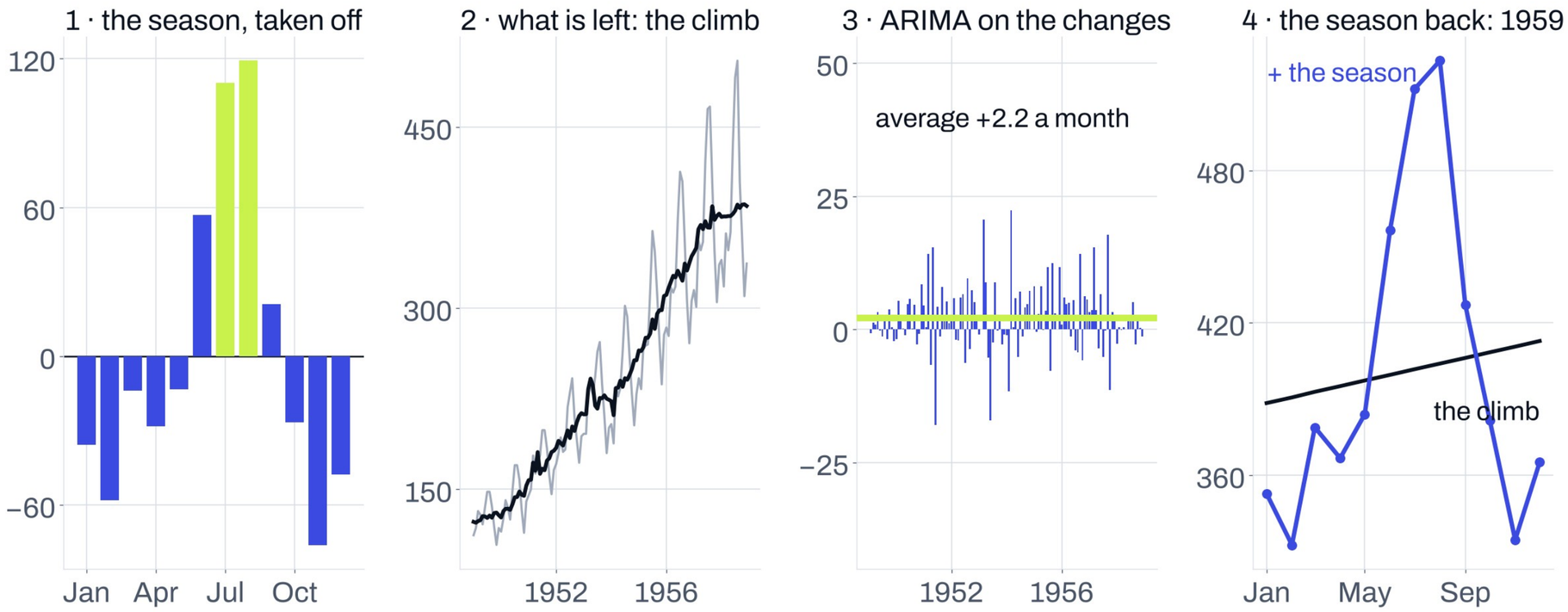
INSTRUCTOR DO

Step 3.5

What ARIMA_PLUS does

Clean the series, find the season, model the trend, add the parts back.

ARIMA_PLUS forecasts the climb, then adds the season back



Grey: the series. Bars: one month's change. ARIMA finds no pattern left in the changes, so the climb goes on straight, about 2.2 a month; then the season goes back on.

STEP 3 · ARIMA(p, d, q), ONE LINE EACH

The trend model in plain words

Part	In words	Our stand-in chose
AR, p	next month leans on the last few months’ values	p = 0
I, d	work with month-to-month changes, not levels	d = 1
MA, q	next month leans on the last few months’ surprises	q = 1
climb	a steady climb added to those changes (Google: drift)	yes

AUTO_ARIMA keeps the order with the lowest AIC, a fit score that charges for every extra term. ML.ARIMA_EVALUATE shows BigQuery’s pick. The season goes back as a fixed amount. What if it grows?

THE RULE TO KEEP

A forecast is scored only on months after its cutoff.

Split by date, and beat the simple forecasts before trusting the model.

PART 03

Judge the forecast

Score 1959 against the simple forecasts, read the interval, then test 1960 once.

EVERYONE DO

Step 4.1

Score the forecast on 1959

Join the forecast to the actual 1959 months and score it like the simple forecasts.

labs/D15/04-Evr-Score-The-Forecast/README.md

STEP 2 · THE SCORE, IN THE SAME UNITS

The forecast against the 1959 months

DATE(f.forecast_timestamp): the forecast's TIMESTAMP becomes a month that joins the DATE column.

```
SELECT ROUND(AVG(ABS(v.passengers - f.forecast_value)), 1) AS mae,
       ROUND(100 * AVG(ABS(v.passengers - f.forecast_value)
                 / v.passengers), 1) AS mape_pct
FROM ML.FORECAST(MODEL ml.air_arima,
                 STRUCT(12 AS horizon, 0.95 AS confidence_level)) AS f
JOIN ml.air_validation AS v
  ON DATE(f.forecast_timestamp) = v.month;
```

→ MAE about 27 · MAPE about 6.2%

The seasonal naive missed by 47.3: the model beats it clearly.

WHAT COMES BACK · STEP 1 · MAY TO OCTOBER 1959 · A STAND-IN

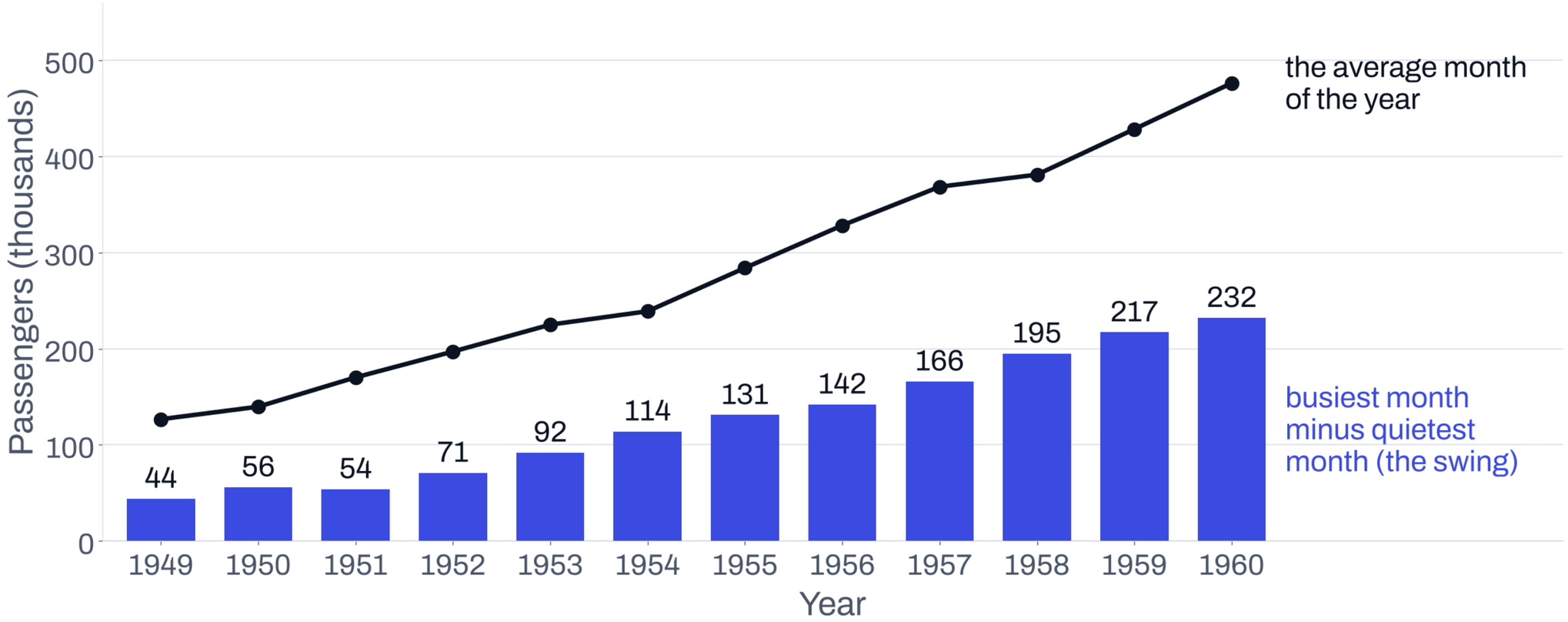
The forecast comes in low all summer

month	passengers	forecast_value	miss
May 1959	420	≈ 384	+36
Jun 1959	472	≈ 456	+16
Jul 1959	548	≈ 512	+36
Aug 1959	559	≈ 523	+36
Sep 1959	463	≈ 427	+36
Oct 1959	407	≈ 382	+25

Every 1959 month comes in above the forecast. MAE averages the twelve misses: about 27.

BUSIEST MONTH MINUS QUIETEST MONTH, YEAR BY YEAR

The summer swing grows as the airline grows



From 44 in 1949 to 232 in 1960. A season added as a fixed amount comes in low in the 1959 summer.

INSTRUCTOR DO

Step 4.2

Reading the interval

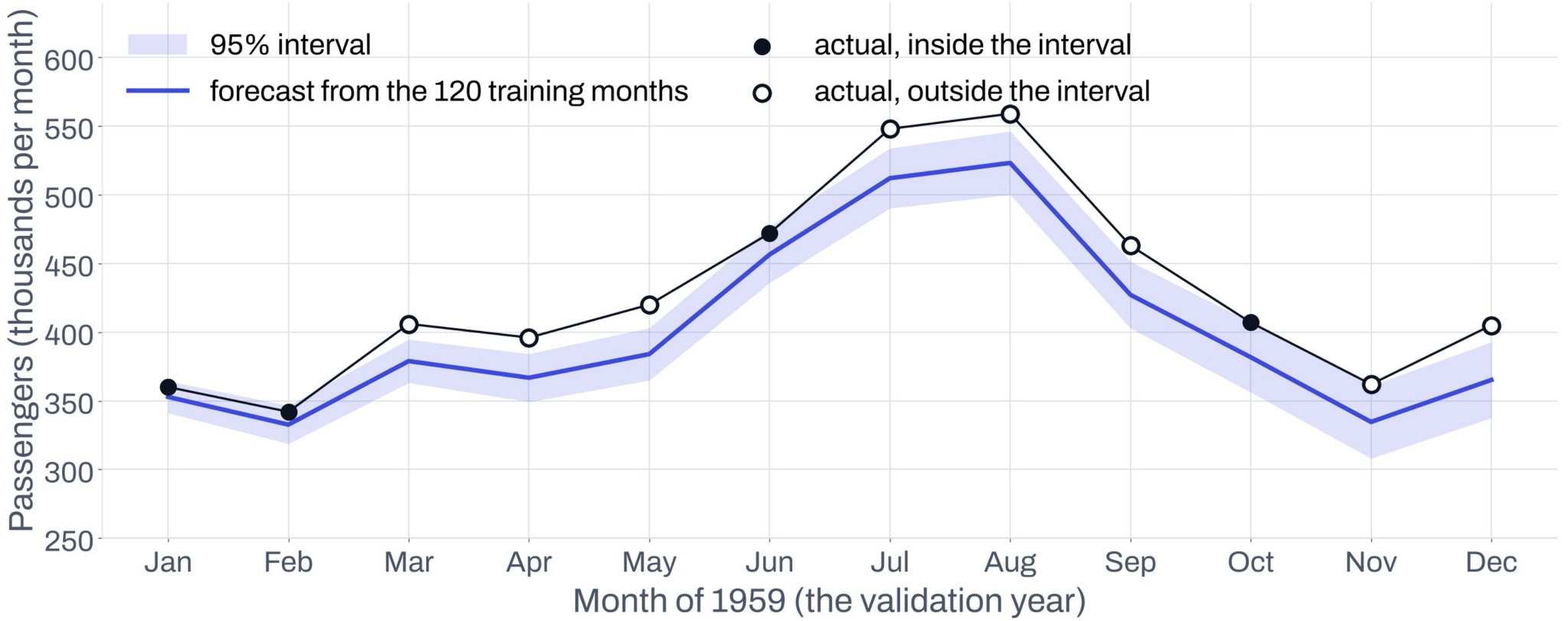
What the range means, why it widens, and which bound a planner uses.

DEFINITION

A 95% prediction interval is the range that should hold one future month’s actual value 95 times in 100. The confidence_interval columns are narrower: they cover the average, not one month.

VALIDATION · 1959 · THE FORECAST AND ITS 95% INTERVAL

A 95% interval that held about 4 of the 12 months



From a Python stand-in that follows ARIMA_PLUS's steps; BigQuery's own interval may hold more.

EVERYONE DO

Step 4.3

Months that look unusual

Every training month split into its parts. What trend and season leave over is where odd months show.

labs/D15/04-Evr-Score-The-Forecast/README.md

STEP 4 · ML.EXPLAIN_FORECAST

Months the parts do not explain

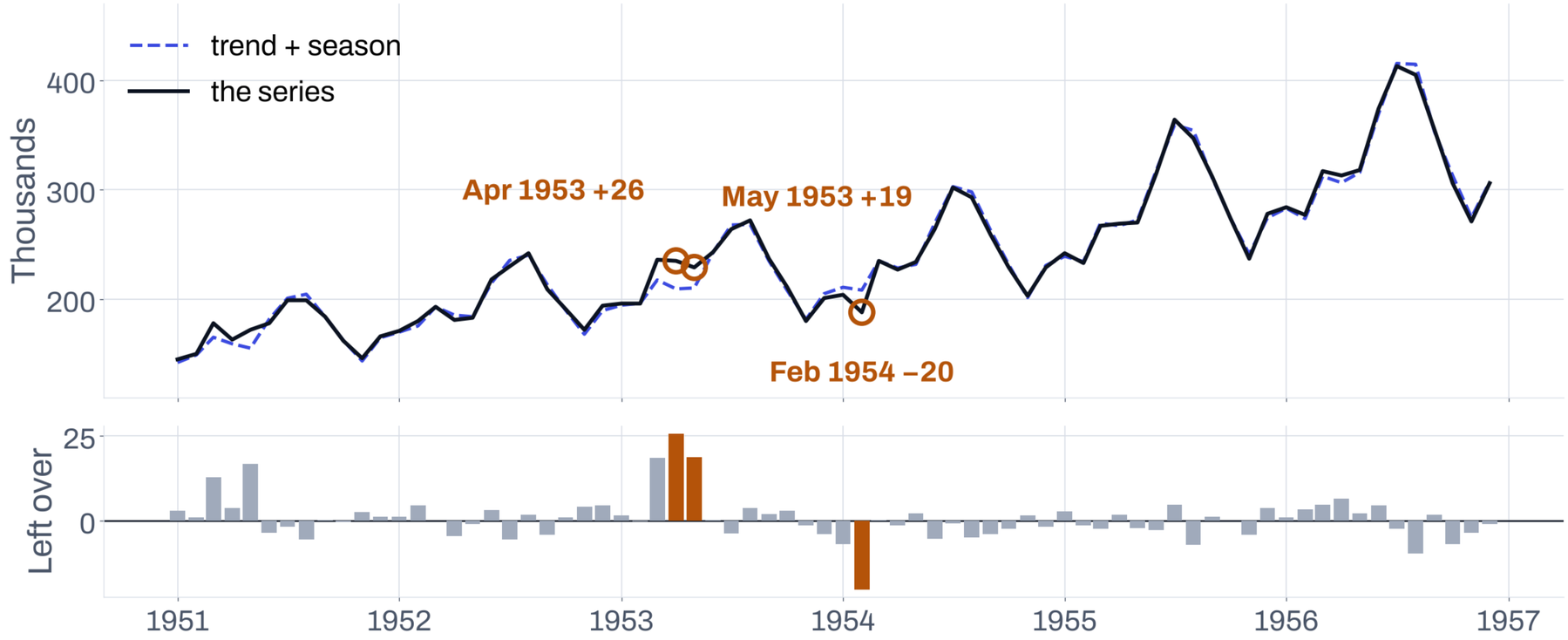
ML.EXPLAIN_FORECAST: every training month split into its parts.

```
SELECT DATE(time_series_timestamp) AS month_history, time_series_data,
       ROUND(trend, 1) AS trend, ROUND(seasonal_period_yearly, 1) AS season,
       spikes_and_dips, ROUND(residual, 1) AS residual
FROM ML.EXPLAIN_FORECAST(MODEL ml.air_arima,
                         STRUCT(12 AS horizon, 0.95 AS confidence_level))
WHERE time_series_type = 'history'
ORDER BY ABS(IFNULL(spikes_and_dips, 0) + IFNULL(residual, 0)) DESC
LIMIT 5;
```

→ about: Apr 1953 +26 · Feb 1954 −20 · May 1953 +19 …

Thousands above or below trend + season; BigQuery calls it residual.

Three months that trend and season do not explain



Amber: the largest leftovers. A flag is a question for the business (a strike, a new route, a typing error), not a row to delete. The model flags; a person decides.

PARTNER DO

Would you plan on this forecast?

Read the 1959 scores and the interval chart. Three sentences for the planner: which forecast, which bound, one risk.

labs/D15/05-Par-Plan-On-The-Forecast/README.md

PARTNER DO

Three sentences for the planner

INSTRUCTIONS	With your partner, use the 1959 scores and the interval chart. Write three sentences: which forecast to plan 1960 on, which bound to staff aircraft and crews to, and one risk. Post them.
HINT	In which months did the model miss most, and in which direction?
BONUS	Why staff to a bound rather than to the forecast itself?

Full steps and Check yourself: labs/D15/05-Par-Plan-On-The-Forecast/README.md

EVERYONE DO

Step 4.5

The test year, once

Retrain on every month before 1960, forecast 1960, and score it one time.

labs/D15/06-Evr-Test-Year-Once/README.md

STEP 1 · THE CHOICE IS MADE

Retrain on 132 months, then score 1960

`WHERE month < '1960-01-01'`: training and validation together, 1949 to 1959.

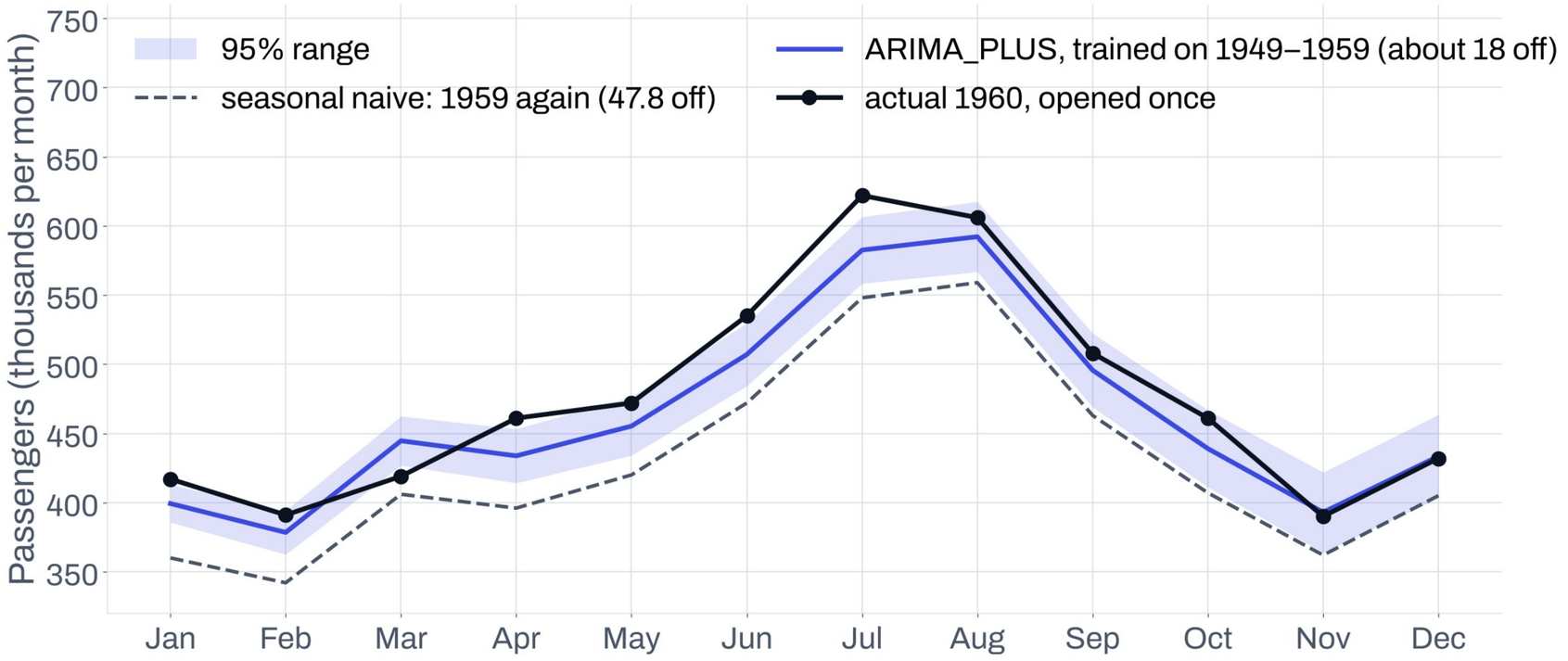
```
CREATE OR REPLACE MODEL ml.air_arima_final
OPTIONS (MODEL_TYPE = 'ARIMA_PLUS',
         TIME_SERIES_TIMESTAMP_COL = 'month',
         TIME_SERIES_DATA_COL = 'passengers',
         HORIZON = 12) AS
SELECT month, passengers FROM ml.air_passengers
WHERE month < '1960-01-01';

                    → 1960: about 18 · seasonal naive from 1959: 47.8
```

Run it once: a second try turns 1960 into another validation year.

TEST · 1960 · SCORED ONCE · A STAND-IN THAT FOLLOWS ARIMA_PLUS

1960, opened once: about 18 off a month, against 47.8



Chosen on 1959, retrained on all 132 months before 1960, then scored once. Change the model after this and 1960 is just another validation year.

REVIEW

Time's up.
Let's review.

Judge the forecast

Solved: labs/D15/05-Par-Plan-On-The-Forecast/Solved/plan_on_the_forecast.md

A SAMPLE ANSWER FOR THE PLANNER

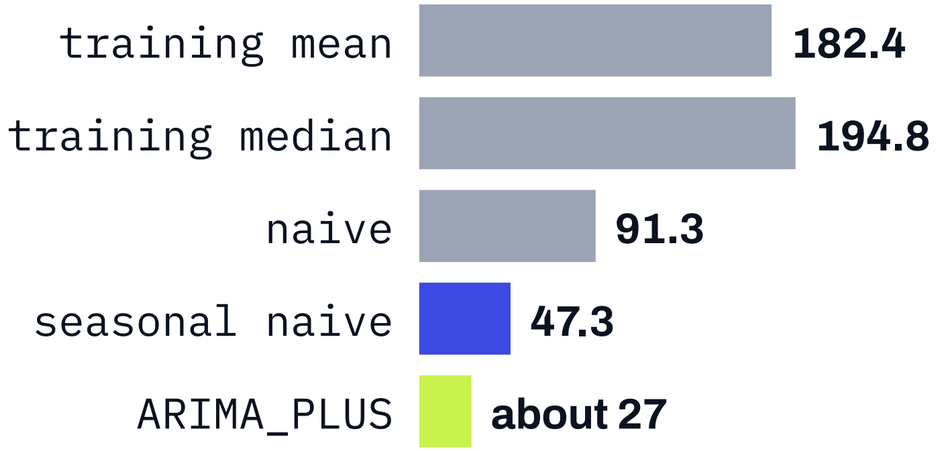
Three sentences

#	Sentence
1	Plan 1960 on ARIMA_PLUS’s forecast: off by about 27 thousand passengers a month on 1959, against 47.3 for last year’s same month.
2	Roster crews to the upper bound from June to September, with a margin: in 1959 the peaks came in above even that.
3	Risk: the season grows with the airline, so the summer peaks come in above the forecast.

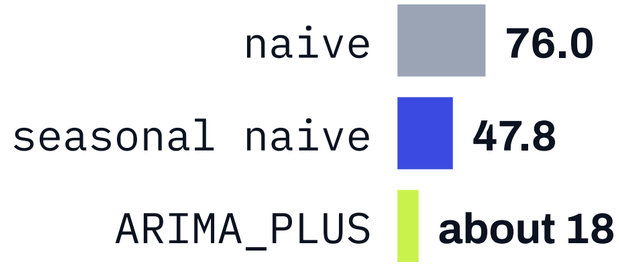
THE SCOREBOARD · MAE, THOUSANDS OF PASSENGERS A MONTH · LOWER IS BETTER

ARIMA_PLUS misses least, in 1959 and again in 1960

1959 · VALIDATION



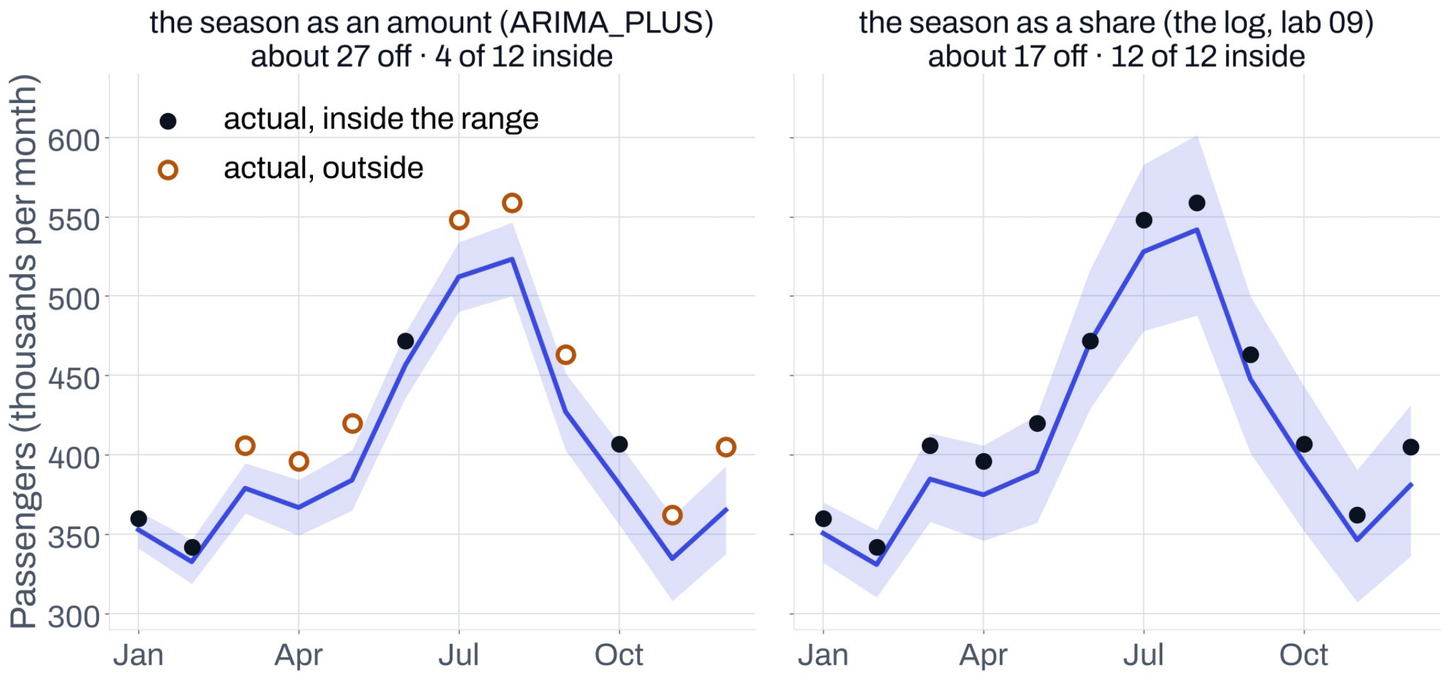
1960 · TEST · ONCE, AFTER THE CHOICE



Plan 1960 on	the seasonal naive: off by 47.3 a month on 1959	×
Plan 1960 on	ARIMA_PLUS: about 27 off on 1959, about 18 on 1960	✓

For 1960 the simple forecasts use 1959’s months. ARIMA_PLUS: the stand-in.

A season that grows with the airline needs a share



ILLUSTRATIVE +10 adds 10 (400 → 410); ×1.1 grows with it (400 → 440). The log makes the season a share: about 17 off, 12 of 12 inside.

THE RULE TO KEEP

A forecast earns trust by beating last year’s same month.

Then read its range: plan to a bound, not to the middle line.

Break

Stretch, then D14’s trees

After the break: why did D14’s trees charge person E so much?

PART 04

Why did the model say that?

One prediction split into pushes, then the model overall. D14’s insurer and its boosted trees.

INSTRUCTOR DO

Step 5.1

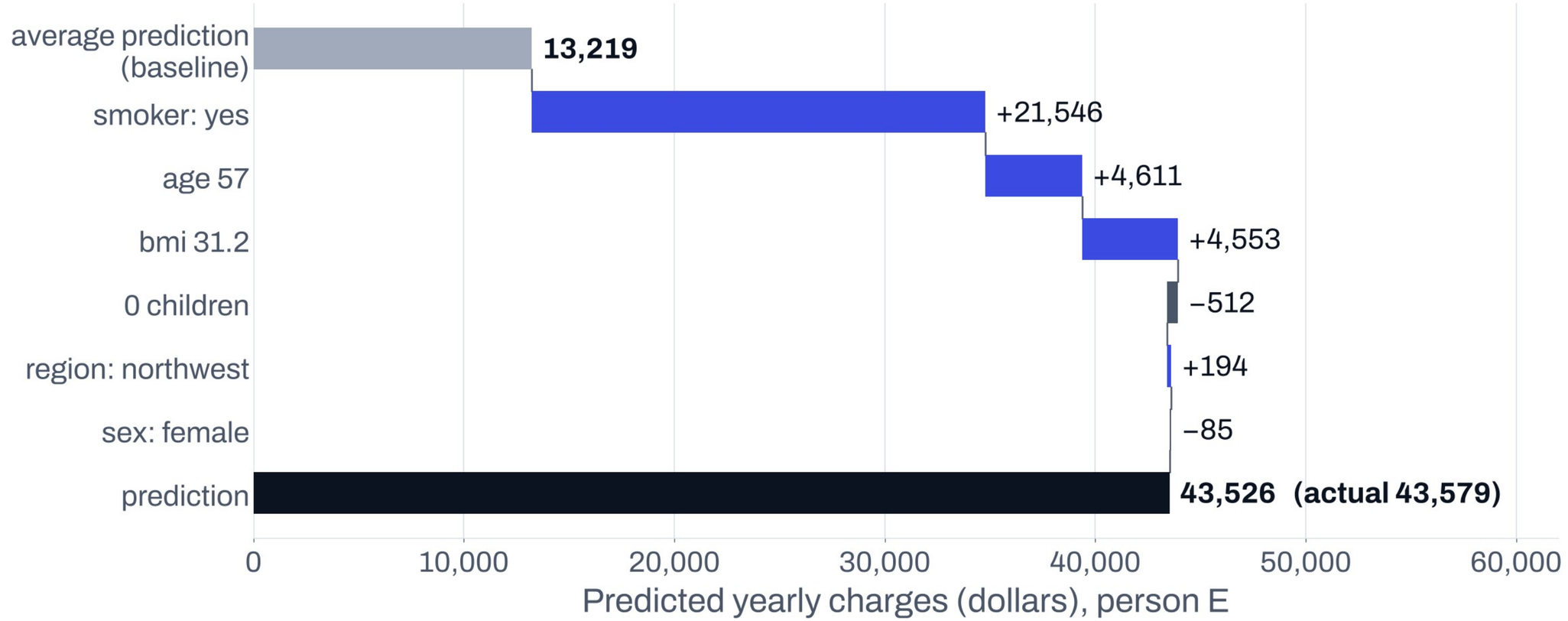
One prediction, many reasons

Person E: 57, female, smoker, BMI 31.2. Predicted about 43,526 dollars; actual 43,579.

An attribution is how far one feature’s value pushed a prediction up or down from the baseline, the model’s average prediction over its training rows.

PERSON E · D14'S BOOSTED TREES · DOLLARS A YEAR

Baseline + pushes = the prediction



About 13,219 + 21,546 + 4,611 + 4,553 − 512 + 194 − 85 = 43,526. Smoking is about half of it.

EVERYONE DO

Step 5.2

Explain five people

ML.EXPLAIN_PREDICT on D14’s boosted model, for people A to E.

labs/D15/07-Evr-Explain-Five-People/README.md

STEP 1 · ONE PREDICTION, SPLIT INTO PUSHES

ML.EXPLAIN_PREDICT: ML.PREDICT plus the reasons

top_k_features: pushes per row, largest first; 6 shows all six.

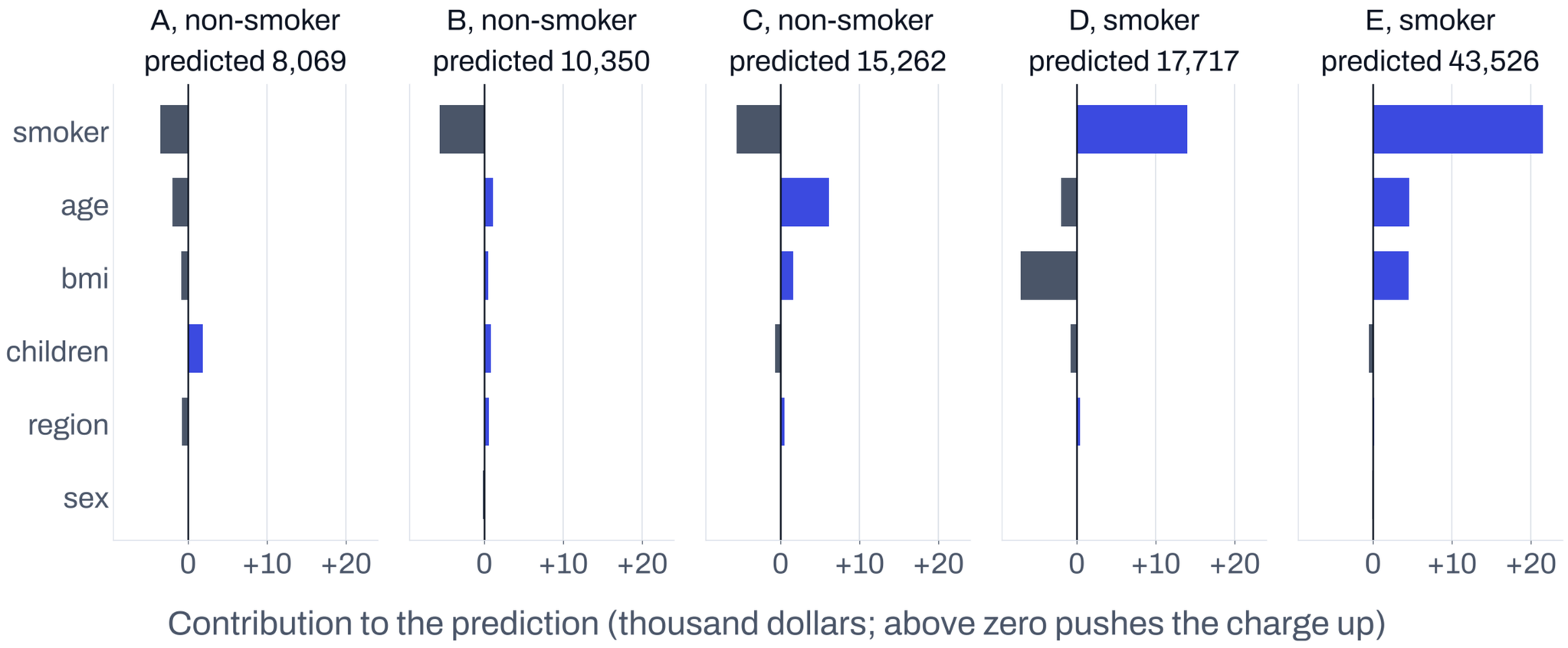
```
SELECT charges, ROUND(predicted_charges) AS prediction,
       ROUND(baseline_prediction_value) AS baseline,
       top_feature_attributions
FROM ML.EXPLAIN_PREDICT(MODEL ml.ins_boosted,
  (SELECT age, sex, bmi, children, smoker, region, charges
   FROM ml.insurance_groups
   WHERE split = 'validation'
     AND ROUND(charges, 2) IN (4391.65, 10370.91, 13228.85, 17904.53, 43578.94)),
  STRUCT(6 AS top_k_features))
ORDER BY charges;
```

→ baseline about 13,219 · E: smoker about +21,546

Trees take 10–15 minutes or more to train. Not ready? labs/D15/Resources/fallback/.

FIVE VALIDATION PEOPLE · ONE BAR PER FEATURE

Smoking pushes D and E up, and A, B and C down



Indigo pushes the predicted charge up, grey pulls it down. Thousands of dollars a year.

INSTRUCTOR DO

Step 5.3

The model overall

Average the size of every push over many people, then compare with D14's feature importance.

STEP 3 · THE GLOBAL EXPLANATION

Switch it on when the model is trained

enable_global_explain = TRUE: ML.GLOBAL_EXPLAIN needs it at training.

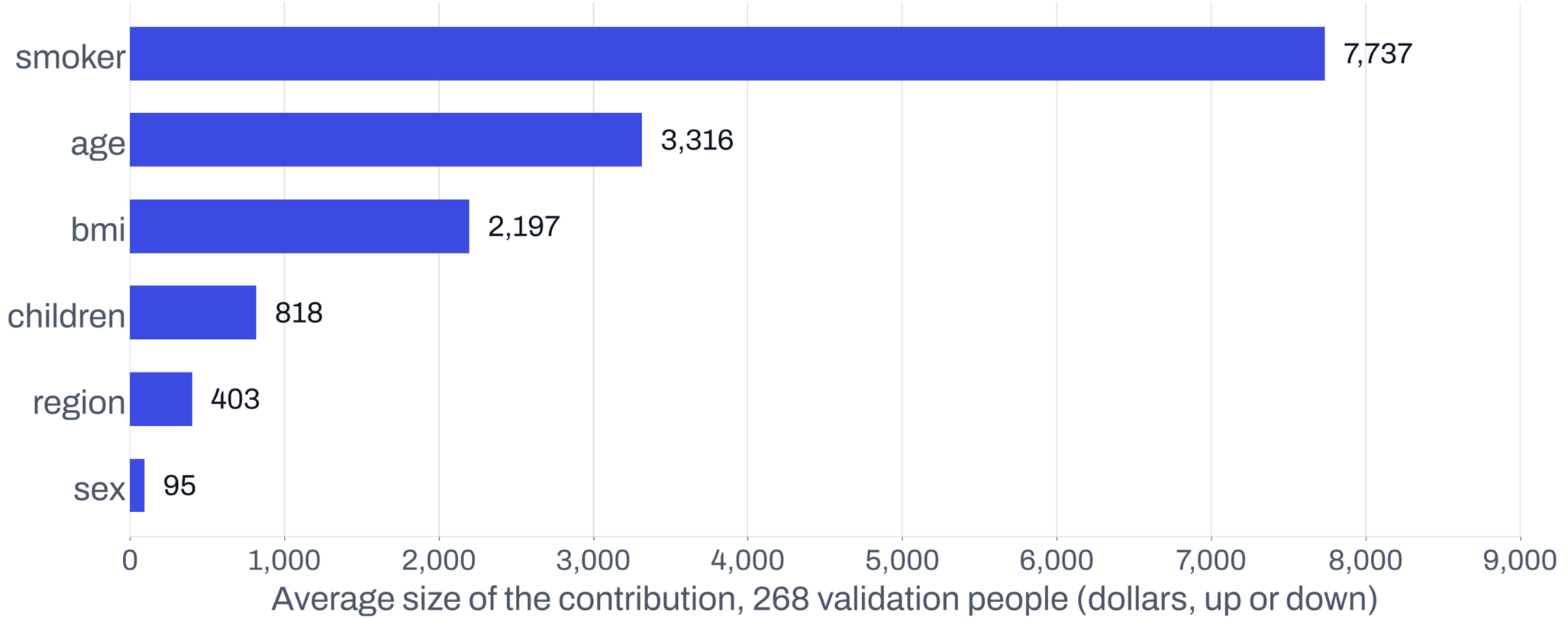
```
CREATE OR REPLACE MODEL ml.ins_explain
OPTIONS (model_type = 'BOOSTED_TREE_REGRESSOR',
         input_label_cols = ['charges'],
         data_split_method = 'NO_SPLIT',
         enable_global_explain = TRUE) AS
SELECT age, sex, bmi, children, smoker, region, charges
FROM ml.insurance_groups WHERE split = 'train';
SELECT feature, ROUND(attribution) AS attribution
FROM ML.GLOBAL_EXPLAIN(MODEL ml.ins_explain)
ORDER BY attribution DESC;

                → smoker about 7,737 · age about 3,316 · bmi about 2,197
```

D14’s trees plus one option. Stand-in numbers: compare the order.

A STAND-IN FOR ML.GLOBAL_EXPLAIN · AVERAGE SIZE OF EACH PUSH · DOLLARS

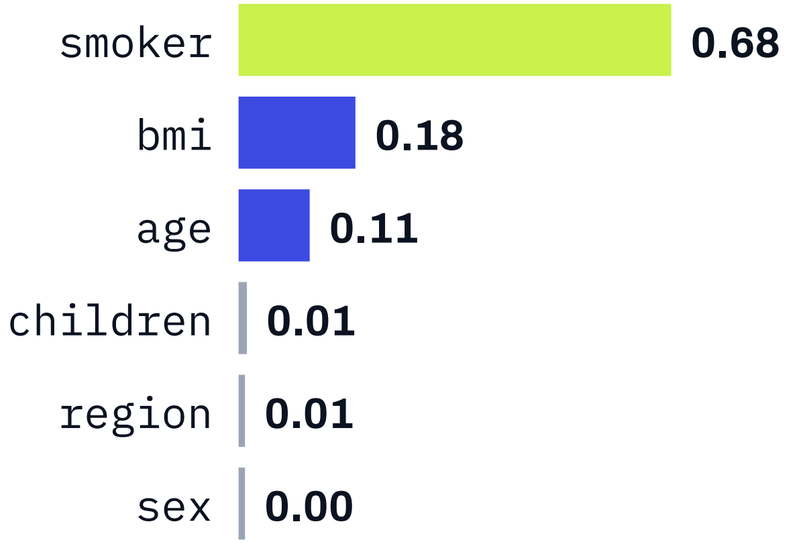
Smoking moves a typical prediction most



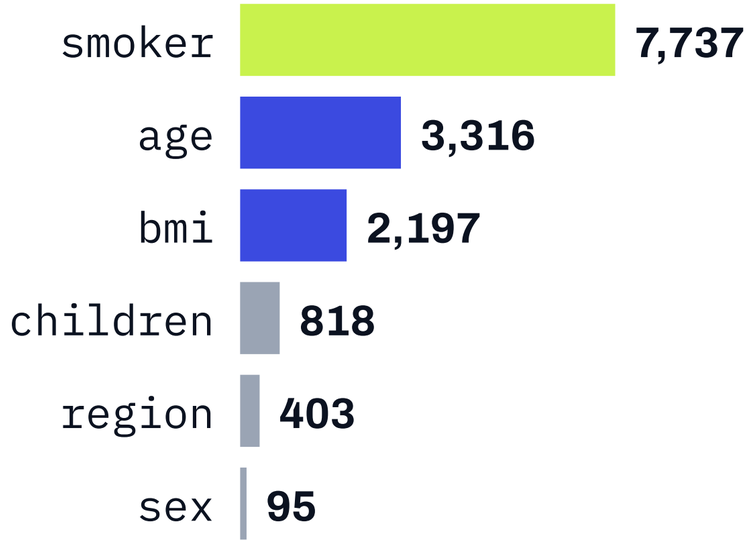
D14’s ML.FEATURE_IMPORTANCE: smoker about 0.68 of all gain. Both describe the model, not what causes charges.

BMI splits the data more; age moves the price more

D14 · ML.FEATURE_IMPORTANCE · share of all gain



D15 · ML.GLOBAL_EXPLAIN · average push, dollars



Gain: how much a feature’s splits helped training (BMI split on 324 times, smoker 20). Push: dollars it moves one prediction, over 268 people.

STUDENT DO

Explain person D to the pricing team

One sentence on why D’s predicted charge is what it is, using his pushes from ML.EXPLAIN_PREDICT.

labs/D15/07-Evr-Explain-Five-People/README.md

STUDENT DO

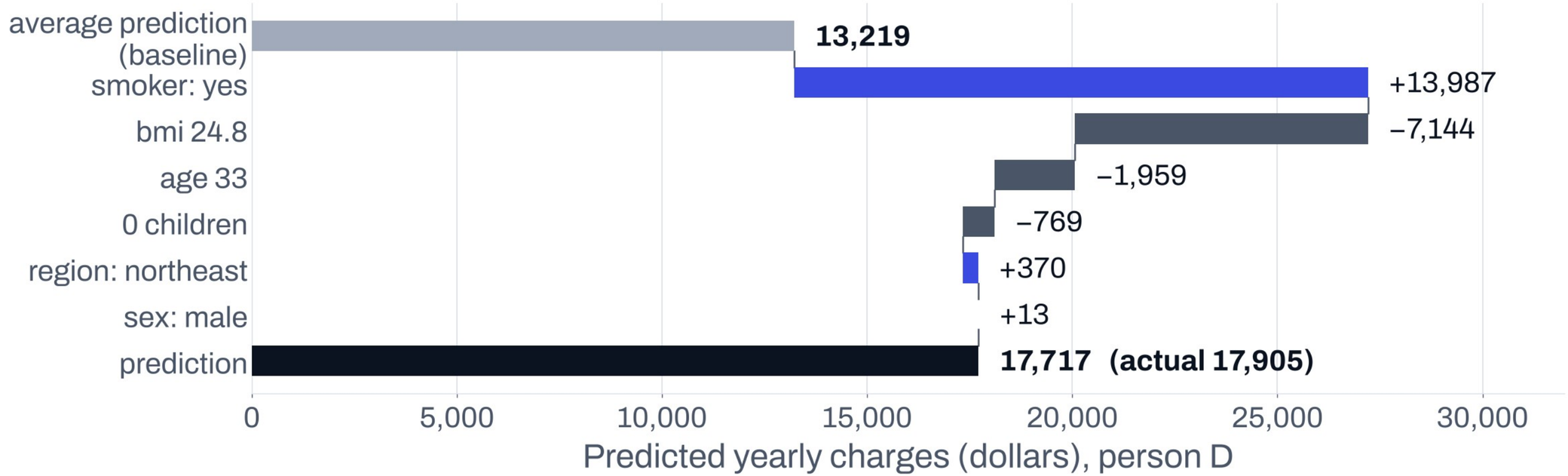
One sentence for the pricing team

INSTRUCTIONS	Person D: 33, male, smoker, BMI 24.8, predicted about 17,717 dollars. Write one sentence a pricing manager can read: what pushed his charge up, what pulled it down, and by how much.
HINT	D is a smoker with a low BMI. Look at his BMI bar.
BONUS	Does BMI push the same way for person E? Why not?

Full steps and Check yourself: labs/D15/07-Evr-Explain-Five-People/README.md

A SAMPLE ANSWER · PERSON D · D14'S BOOSTED TREES · DOLLARS A YEAR

Smoking adds; a low BMI takes some back



"Smoking adds about 14,000 dollars to D's charge; his BMI of about 25 takes about 7,100 back." E's higher BMI, 31.2, adds about 4,553 to hers.

PART 05

Fix the leak, then serve

Cut points from the training rows only, then two tables Power BI can read.

INSTRUCTOR DO

Step 6.1

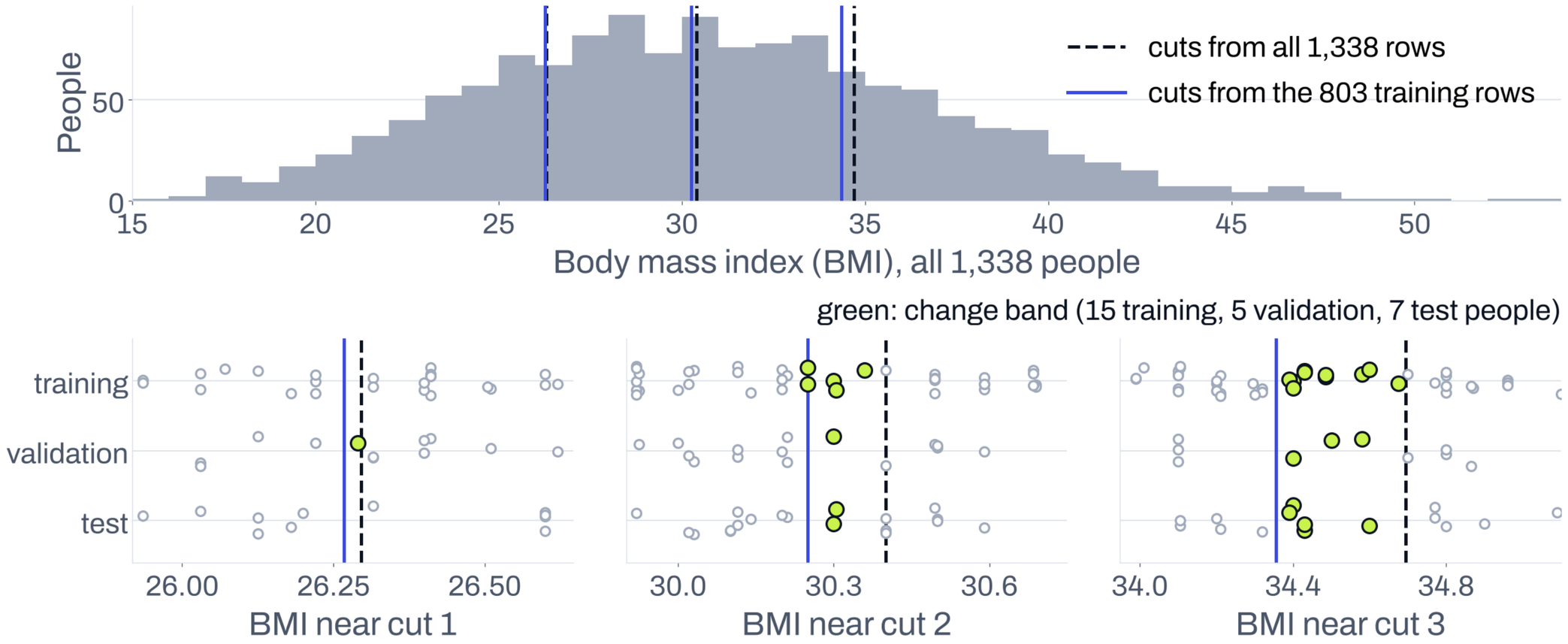
TRANSFORM keeps the training rules

D14 lab 10 cut BMI into four bands using every person, test people included.

labs/D15/08-Evr-Transform-And-Save/README.md

BMI QUARTILE CUTS · ALL 1,338 PEOPLE AGAINST THE 803 TRAINING PEOPLE

The test people helped choose the cut points



All rows: 26.30 · 30.40 · 34.69. Training only: 26.27 · 30.25 · 34.36. 5 validation, 7 test people move.

DEFINITION

TRANSFORM puts the preprocessing inside the model: the cut points are computed from the training rows and reused, unchanged, at every prediction.

STEP 1 · PREPROCESSING INSIDE THE MODEL

Four BMI bands, cut on the training rows

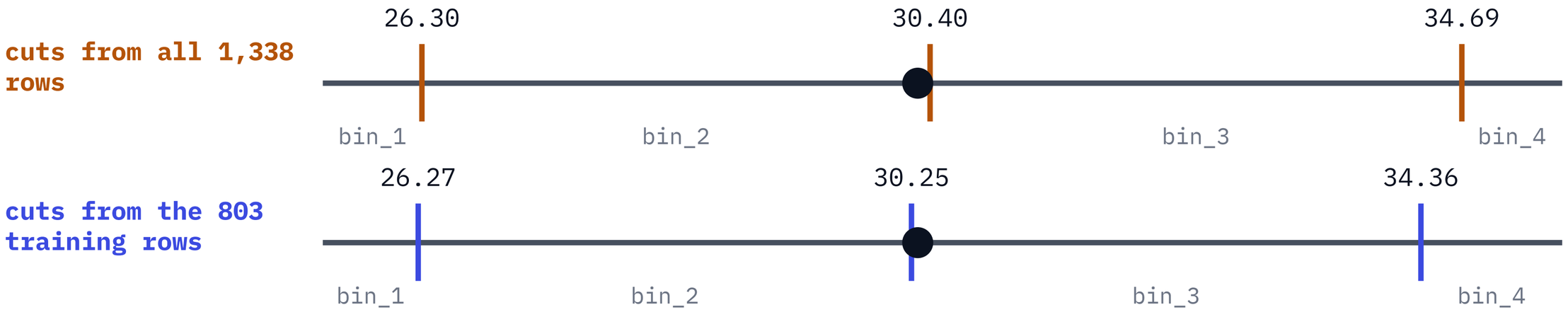
TRANSFORM sees only the rows the SELECT gives it: the 803 training people.

```
CREATE OR REPLACE MODEL ml.ins_transform
TRANSFORM (ML.QUANTILE_BUCKETIZE(bmi, 4) OVER () AS bmi_band,
           age, sex, children, smoker, region, charges)
OPTIONS (model_type = 'LINEAR_REG',
         input_label_cols = ['charges'],
         data_split_method = 'NO_SPLIT') AS
SELECT age, sex, bmi, children, smoker, region, charges
FROM ml.insurance_groups
WHERE split = 'train';
```

Google: “the same statistics are automatically applied to the input in prediction.”

ONE VALIDATION PERSON · ROW 288 · 27, male, BMI 30.30

With TRANSFORM, a person gets the band training chose



No TRANSFORM	the test people helped choose the cuts: this person is in bin_2	×
TRANSFORM	cuts from training, kept in the model: bin_3 at every prediction	✓

The dot is BMI 30.30. 5 validation and 7 test people change band like this.

THE RULE TO KEEP

Anything computed from rows comes from the training rows.

Cut points, averages, scales: TRANSFORM keeps them with the model.

EVERYONE DO

Step 6.2

Save predictions where Power BI can read them

Two ml.* tables: the 1960 forecast and every insured person scored.

labs/D15/08-Evr-Transform-And-Save/README.md

STEP 2 · A TABLE FOR THE REPORT

The 1960 forecast, saved as a table

CREATE OR REPLACE TABLE: the sandbox runs no INSERT, so one statement writes the table.

```
CREATE OR REPLACE TABLE ml.air_forecast AS
SELECT DATE(f.forecast_timestamp) AS month,
       ROUND(f.forecast_value, 1) AS forecast_passengers,
       ROUND(f.prediction_interval_lower_bound, 1) AS lower_95,
       ROUND(f.prediction_interval_upper_bound, 1) AS upper_95,
       te.passengers AS actual_passengers
FROM ML.FORECAST(MODEL ml.air_arima_final,
                 STRUCT(12 AS horizon, 0.95 AS confidence_level)) AS f
LEFT JOIN ml.air_test AS te ON te.month = DATE(f.forecast_timestamp);

                                  → 12 rows: 1960, forecast beside actual
```

Then ml.insurance_scored, 1,338 rows; no table shares a column’s name.

WHAT COMES BACK · ml.air_forecast · THE FIRST SIX OF 12 ROWS · A STAND-IN

One row per month: the forecast, its range, the actual

month	forecast_passengers	lower_95	upper_95	actual_passengers
1960-01-01	≈ 399.3	≈ 385.7	≈ 412.9	417
1960-02-01	≈ 378.4	≈ 362.6	≈ 394.3	391
1960-03-01	≈ 444.6	≈ 426.8	≈ 462.4	419
1960-04-01	≈ 433.8	≈ 414.2	≈ 453.3	461
1960-05-01	≈ 455.3	≈ 434.1	≈ 476.4	472
1960-06-01	≈ 507.0	≈ 484.4	≈ 529.7	535

A report can draw the line, shade the range and score the year without running the model again.

BigQuery ML functions today

In class	What it gives you	Activity
ML.FORECAST	the next months, each with a lower and upper bound	02, 04–06, 08
ML.EXPLAIN_FORECAST	each month split into trend, season, spikes and dips	04
ML.EXPLAIN_PREDICT	each prediction with the features that pushed it	07
ML.GLOBAL_EXPLAIN	how much each feature drives predictions, on average	07
ML.FEATURE_IMPORTANCE	trees only: how often and how usefully each feature split	07
ML.QUANTILE_BUCKETIZE	bands of about equal size; TRANSFORM keeps the cut points	08
ML.PREDICT	a prediction for each row you pass in	08

Lab	New in the labs
09, 10, 13	**ML.ARIMA_EVALUATE · ML.ARIMA_COEFFICIENTS**
10	**ML.DETECT_ANOMALIES (billed, even in a sandbox) · ML.EVALUATE · ML.METRICS (Preview)**
12	**ML.TRANSFORM · ML.FEATURE_INFO · ML.BUCKETIZE · ML.FEATURE_CROSS · ML.POLYNOMIAL_EXPAND**
12	**ML.STANDARD_SCALER · ML.MIN_MAX_SCALER · ML.MAX_ABS_SCALER · ML.ROBUST_SCALER**
12	**ML.ONE_HOT_ENCODER · ML.LABEL_ENCODER · ML.IMPUTER · more in the lab’s table**
14	**ML.GET_INSIGHTS, on a CONTRIBUTION_ANALYSIS model**

CLOSE

What you can do now

- Forecast a series; read the interval
- Split by date; beat simple forecasts
- Explain a prediction and the model
- Fix a leak with TRANSFORM; save ml.* tables

NEXT · D16 Explaining again, MLOps, four models, Power BI.

AFTER CLASS · SELF-GUIDED · NOT SUBMITTED

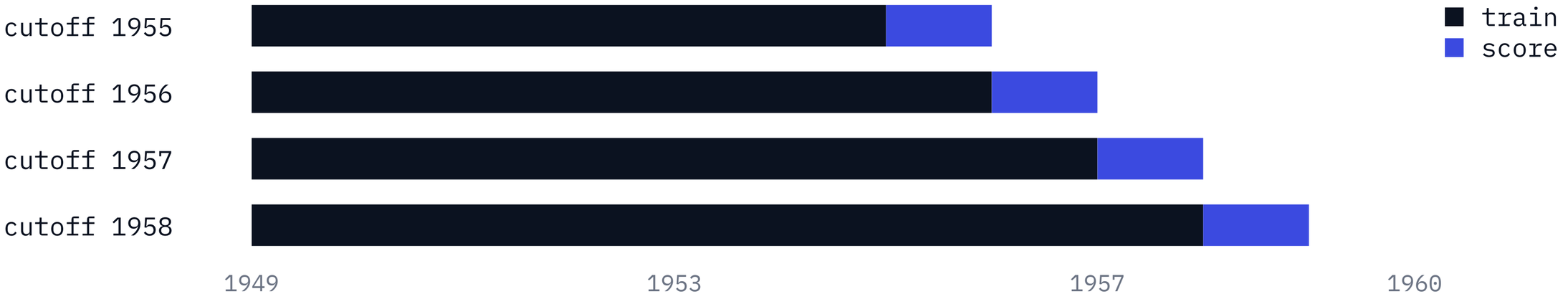
Labs 09 to 14, option by option

Lab	Folder in labs/D15/	What you try
09	09-Lab-Arima-Plus-Options	every CREATE MODEL option, one per step; checkpoints C1–C3; the log fix
10	10-Lab-Forecast-Functions	ML.FORECAST, ML.EXPLAIN_FORECAST, ML.ARIMA_EVALUATE; ML.DETECT_ANOMALIES, billed even in the sandbox
11	11-Lab-Explain-Options	ML.EXPLAIN_PREDICT and ML.GLOBAL_EXPLAIN options, beside ML.FEATURE_IMPORTANCE
12	12-Lab-Transform-Functions	TRANSFORM with bucketing and scaling; ML.TRANSFORM shows the features
13	13-Lab-Nakheel-Daily-Revenue	optional: Nakheel’s daily revenue, forecast and saved for D22
14	14-Lab-Nakheel-Revenue-Drivers	optional: which slices explain Nakheel’s revenue change (ML.GET_INSIGHTS)

APPENDIX · BEYOND TODAY

One test year is one check; there are more

SCORE SEVERAL YEARS · TRAIN BEFORE EACH CUTOFF, SCORE THE NEXT 12 MONTHS



ARIMA_PLUS_XREG

plus columns that move with the series; you give their future values

AI.FORECAST

Google’s trained TimesFM, no CREATE MODEL; Preview in October 2026

Winning every year beats winning once.